Corrections and resource bounds for the arXiv v2 Trotter transport algorithm

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Abstract

We audit the supplied arXiv v2 of *Trotter-based quantum algorithm for solving transport equations with exponentially fewer time-steps* by Zylberman, Fredon, Loureiro and Debbasch. Exact counterexamples show that its conservation claim for spatial derivative norms is false for variable transport fields, its displayed controlled-SWAP circuit measures squared overlap rather than signed overlap, and several formulas have inconsistent signs, normalization factors, or index ranges. We provide corrections, a finite-time regularity estimate, and a coefficient-aware finite-grid commutator bound that scales as the inverse mesh spacing rather than its square for smooth periodic coefficients. This sharpens the operator-norm comparison underlying the advertised exponential improvement without disproving the value of smooth-state analysis. Independent calculations reproduce the reported eight-point discretization errors and exhibit a separate smooth periodic example whose fixed-step splitting error stabilizes under refinement. We distinguish literal equation failures, resource qualifications and unavailable implementation evidence. The results concern arXiv:2508.15691v2, whose PDF and source were verified; the subsequent journal version was identified but its full text was inaccessible and is not audited here. No end-to-end quantum advantage or general impossibility claim is made.

1. Introduction

Encoding a discretized transport solution in quantum amplitudes can reduce the number of memory qubits dramatically. The remaining questions concern how that state is prepared, how accurately it evolves, what measurements return, and what resources each of these operations consumes. A statement about a time discretization bound answers only one of those questions. It should be judged on its own assumptions before being interpreted as a computational advantage over a classical solver.

The audited preprint [1] studies periodic transport with a velocity field whose component in each coordinate direction is independent of that same coordinate. It combines centered finite differences, quantum Fourier transforms and first-order product formulas. Its central analytical observation is sensible: an operator containing an increasingly fine discrete derivative can have a large worst-case norm while acting moderately on a fixed, smooth function. The paper compares a vector-dependent error estimate with a crude product of operator norms and reports an exponential reduction in the predicted number of steps as the number of qubits per coordinate grows.

Several details require correction before this conclusion can be used reliably. Smoothness is preserved for a finite time under a smooth transport flow, but its derivative norms need not retain their initial values. Two separately prepared states do not supply a phase reference through a SWAP test. A sign convention for a shift acting on ket labels has the opposite action on its amplitude array. These issues are algebraic and can be checked without access to a quantum computer. They are also distinct from whether the program unavailable in the inspected arXiv materials used the printed expressions literally.

This paper has three purposes. First, we identify reproducible failures in specified equations of the supplied edition. Second, we replace the incorrect derivative conservation argument and improve

the operator-norm baseline using the structure of finite differences. Third, we delimit what survives: unitary directional transport, the practical benefit of smooth-state estimates, and the reported small-grid error magnitudes. This separation prevents a transcription error from being mistaken for a general impossibility result and prevents a valid numerical illustration from repairing an invalid proof step automatically.

Our contribution is a version-specific technical comment, not an authorized revision of the original authors' paper. A later article with the same title appeared in *Quantum Science and Technology* 11, 025015 (2026), DOI 10.1088/2058-9565/ae488e [2]. The publisher required a human access challenge in the available sessions. We therefore do not assert that any identified arXiv error remains in that article. The version boundary is part of the result, not a footnote to be removed when citing this audit.

2. Related work and mathematical setting

An, Fang and Lin [3] established the usefulness of vector-norm analysis for time-dependent Hamiltonian simulation with unbounded operators. Their treatment includes an unbounded kinetic operator and a bounded potential under stated commutator assumptions. The audited paper acknowledges that antecedent and addresses a different transport decomposition. Our counterexamples do not dispute the earlier theorem, nor do they establish that the transport paper has no new analytical content. They address specific claims in its extension and the interpretation of its comparison baseline.

The controlled-SWAP identity is established in the quantum fingerprinting literature [4]. Hadamard tests and quantum amplitude estimation [5] can estimate other quantities with suitable controlled preparation and access assumptions. These protocols are not interchangeable solely because each produces a scalar measurement. In particular, retaining a phase reference through a controlled preparation differs physically from comparing two independently prepared state registers.

Throughout, the spatial domain is the unit torus, with smooth periodic real velocity $c = (c_1, \ldots, c_d)$, and

$$\partial_t f + c \cdot \nabla f = 0. \tag{1}$$

We adopt the original componentwise condition $\partial_j c_j = 0$ when discussing its directional implementation. This implies $\nabla \cdot c = 0$, so the continuum flow preserves volume and the solution's L^2 norm. It is stronger than divergence freedom. Unless stated otherwise, dimension, finite-difference order, final time and continuum input functions remain fixed while the grid is refined. Constants may depend on all of those fixed quantities.

Let $h_j = 1/N_j$, $N_j = 2^{n_j}$, and a_k be the centered first-derivative coefficients of order $2p$, with $a_1 = 1/2$ for $p = 1$. We distinguish a forward ket shift $S_j|x\rangle = |x + h_j e_j\rangle$ from a forward amplitude translation $\tau_{j,k} g(x) = g(x + k h_j e_j)$. They are inverse conventions. The consistent amplitude derivative used below is

$$B_j = \frac{1}{h_j} \sum_{k=-p}^{p} a_k \tau_{j,k}, \qquad B_j^\dagger = -B_j. \tag{2}$$

The Hermitian transport Hamiltonian corresponding to (1) is $H = -i \sum_j C_j B_j$, because $-iH = -\sum_j C_j B_j$. Here C_j is the diagonal matrix of samples of c_j. Under the componentwise condition, C_j and B_j commute. This sign convention will be kept explicit in every test.

3. Methodology and evidence boundary

The supplied PDF has 34 pages, 4,798,118 bytes and SHA-256 `8c3e5bf0c6c063bb96f383066c8a5640659e6a69a40a4d990e90901dd56d0fd2`. It is byte-identical to the official versioned arXiv v2 PDF. We read its full text and appendices, inspected the displayed equations, and compared them against the official TeX archive. Equation and page locators below refer exclusively to that edition. The source archive contains the manuscript, bibliography and figure images; it does not contain the original study's simulation programs or raw hardware outcomes. An absence in this inspected archive is not a claim that no code exists anywhere.

We use two independent calculation scripts. `experiments/reproduce.py` generates analytical counterexample samples, finite-matrix checks, five CSV tables and three figures. `experiments/resource_counterchecks.py` independently checks the main measurement and convention examples, reproduces the reported hardware-discretization error magnitudes, and evolves a separate smooth periodic transport problem. Both use double-precision NumPy; Matplotlib is required only for the first script's figures. There is no fitting to experimental data, stochastic selection of favorable runs, or use of a local language model as a numerical oracle. The calculations use CPU arrays; releasing GPU memory is not a condition for these small checks.

We classify evidence at three levels. A literal equation failure is demonstrated by substitution or a finite matrix whose result contradicts the printed statement. A resource qualification identifies parameters hidden when an asymptotic assertion fixes an input family. An implementation uncertainty identifies what cannot be inferred from manuscript formulas without the actual program or data. The first level justifies correction; the second justifies narrower wording; the third justifies restraint rather than an accusation.

The main numerical tolerances are 10^{-11} for analytic shear quadrature and 10^{-12} for elementary SWAP identities. The translation test refines N from 16 to 128. The commutator test uses $N = 8, 16, 32, 64, 128$ per coordinate, exploiting a tensor factorization to evaluate the exact operator norm using only N-dimensional matrices. For the separate periodic evolution diagnostic, a 2,048-step Strang reference is compared with a 1,024-step reference to quantify reference error. That difference is reported alongside the first-order splitting error rather than assumed negligible silently.

The repository supplies source provenance, a correction registry, numerical outputs, environment versions, and a hash manifest. The tests are deliberately small and falsifiable. They verify algebra and illustrate limiting behavior; they are not a hardware replication, an assessment of fault-tolerant gate counts, or a benchmark against an optimized classical PDE code. We have not attempted to infer the content of the inaccessible journal text from its title or metadata.

4. Results: regularity and finite-grid resource bounds

4.1. Spatial derivatives are forced by velocity gradients

The paragraph following Theorem III.4 on p.12, its footnote 10, and the opening of Appendix B on p.29 assert conservation of the L^2 norm of every spatial derivative. The appendix argues that each derivative solves the original homogeneous transport equation. For variable c, differentiation gives instead

$$(\partial_t + c \cdot \nabla)\partial_\ell f = -\sum_j (\partial_\ell c_j)\partial_j f. \tag{3}$$

The componentwise constraint removes $\partial_j c_j$, not every cross derivative. Thus it does not remove the right-hand side of (3). Volume preservation conserves $\|f\|_2$; it does not remove deformation of level sets or amplification of gradients.

An exact smooth periodic counterexample satisfying every componentwise condition is

$$c(x,y) = (\sin(2\pi y), 0), \quad f_0(x,y) = \sin(2\pi x), \quad f_t(x,y) = \sin(2\pi[x - t\sin(2\pi y)]). \tag{4}$$

Direct integration gives

$$\|f_t\|_2 = 2^{-1/2}, \qquad \|\partial_x f_t\|_2 = \sqrt{2}\pi, \qquad \|\partial_y f_t\|_2 = 2\pi^2|t|. \tag{5}$$

The last norm grows from zero to 19.7392088 at time one. This is a continuum counterexample, not an aliasing effect. The sampled quadrature agrees with (5) to the stated tolerance. A nonnegative version follows by adding a constant to f_0; the derivative contradiction remains unchanged. Consequently, interpreting the encoded function as a nonnegative density does not rescue the conservation claim.

There is a constructive replacement. Let Φ_t be the flow map of c, so $f_t(\Phi_t(a)) = f_0(a)$. Differentiating with respect to a yields

$$\nabla f_t(\Phi_t(a)) = D\Phi_t(a)^{-T} \nabla f_0(a). \tag{6}$$

Since $\det D\Phi_t = 1$ for a divergence-free field, the usual flow estimate implies

$$\|\nabla f_t\|_2 \le \exp\left(\int_0^t \|\nabla c(\cdot, s)\|_{\mathrm{op},\infty}\, ds \right) \|\nabla f_0\|_2. \tag{7}$$

Equation (7) bounds the entire gradient, not each component by its own initial component. It therefore permits a zero initial y derivative to become nonzero. For integer $s \ge 1$, the multi-index identity

$$(\partial_t + c\cdot\nabla) D^\alpha f = -\sum_{0<\beta\le\alpha} \binom{\alpha}{\beta} (D^\beta c)\cdot \nabla D^{\alpha-\beta} f \tag{8}$$

gives an H^s energy bound with a factor $\exp(C_{s,d}\int_0^t \|\nabla c\|_{W^{s-1,\infty}} ds)$, under integrable coefficient bounds and periodic sufficiently regular data. This follows by multiplying differentiated equations by $D^\alpha f$, integrating, using divergence freedom on the transport term, and applying Cauchy-Schwarz and Gronwall. It supplies finite-time regularity constants without asserting derivative conservation.

The theorem's displayed leading coefficient already retains a maximum over time of derivative norms. That form can survive this correction. The invalid step is replacing those norms with their initial values and describing all time dependence by an explicit factor T^2. A defensible estimate keeps K_T and the time-dependent derivative ratios visible. It can remain independent of mesh spacing for fixed smooth inputs and finite time while depending strongly on final time, velocity gradients, localization, and dimension. We do not prove the full original theorem with every numerical constant by this replacement alone.

4.2. A coefficient-aware operator comparison

The paper obtains its $O(4^n)$ comparison from bounding a commutator by a product of two derivative operator norms, each $O(N)$ for $N = 2^n$ grid points per coordinate. Smooth coefficients permit cancellation before any state-dependent argument is introduced. Define

$$A_p = \sum_{k=-p}^{p} |a_k|, \qquad M_p = \sum_{k=-p}^{p} |k a_k|. \tag{9}$$

Periodic Lipschitz continuity gives the exact finite-grid identity and bound

$$[B_j, C_m] = \sum_k \frac{a_k}{h_j} [c_m(x + k h_j e_j) - c_m(x)] \tau_{j,k}, \qquad \|[B_j, C_m]\| \le M_p \|\partial_j c_m\|_\infty. \tag{10}$$

This follows term by term from the mean value estimate; translations are unitary. It requires coefficients smooth across the periodic seam. A discontinuous periodized linear coordinate function does not satisfy that hypothesis globally.

Since the coordinate differences commute,

$$[C_j B_j, C_m B_m] = C_j [B_j, C_m] B_m - C_m [B_m, C_j] B_j. \tag{11}$$

Combining $\|B_j\| \le A_p / h_j$ with (10) gives

$$\|[C_j B_j, C_m B_m]\| \le A_p M_p \left(\frac{\|c_j\|_\infty \|\partial_j c_m\|_\infty}{h_m} + \frac{\|c_m\|_\infty \|\partial_m c_j\|_\infty}{h_j} \right). \tag{12}$$

For fixed order and balanced meshes, (12) is $O(N)$ rather than $O(N^2)$. For autonomous skew-Hermitian directional generators, the double-integral Duhamel representation of first-order Lie splitting bounds a local defect by $\Delta t^2/2$ times the sum of pairwise commutator norms. Unitarity lets local errors telescope without norm amplification. The global bound is therefore $T^2/(2L)$ times that sum. This standard argument applies directly to (12); it does not require estimating a derivative norm of the state. Endpoint freezing of a time-dependent coefficient adds a term controlled by its time derivative and $\|B_j\|$, again $O(N)$ under the corresponding fixed bounds. We restrict the constructive splitting statement here to the autonomous case to avoid claiming a full new time-dependent theorem.

For $c_1(y) = \sin(2\pi y)$, $c_2 = 1$ and $p = 2$, the commutator factorizes. The exact operator norm grows from 67.8300 at $N = 8$ to 1782.5930 at $N = 128$. The crude bound grows from 227.5556 to 61676.6654, while (12) gives 125.6637 and 2010.6193. On the fixed smooth state proportional to $\sin(2\pi x)$, the commutator norm approaches 27.9154568. These values separate three distinct quantities: a crude worst-case bound, a sharper worst-case bound, and action on a chosen smooth state.

Equation (12) does not invalidate the reported factor relative to the specific $O(N^2)$ baseline. It shows that the factor is not a lower bound on how many steps operator-norm methods require. Relative to this sharper baseline the displayed leading comparison is $N = 2^n$, still exponential in qubits, rather than $N^2 = 4^n$. Neither comparison establishes an end-to-end quantum/classical speedup. No novelty priority is claimed for the elementary commutator identity itself; its role here is to make the comparison in this particular audit explicit.

4.3. The asymptotic regime must couple space and time refinement

Appendix B retains remainders of the form

$$R_q = O\left(\frac{T^q}{L^{q-1} h^q} \right), \quad q = 3, 4, 5. \tag{13}$$

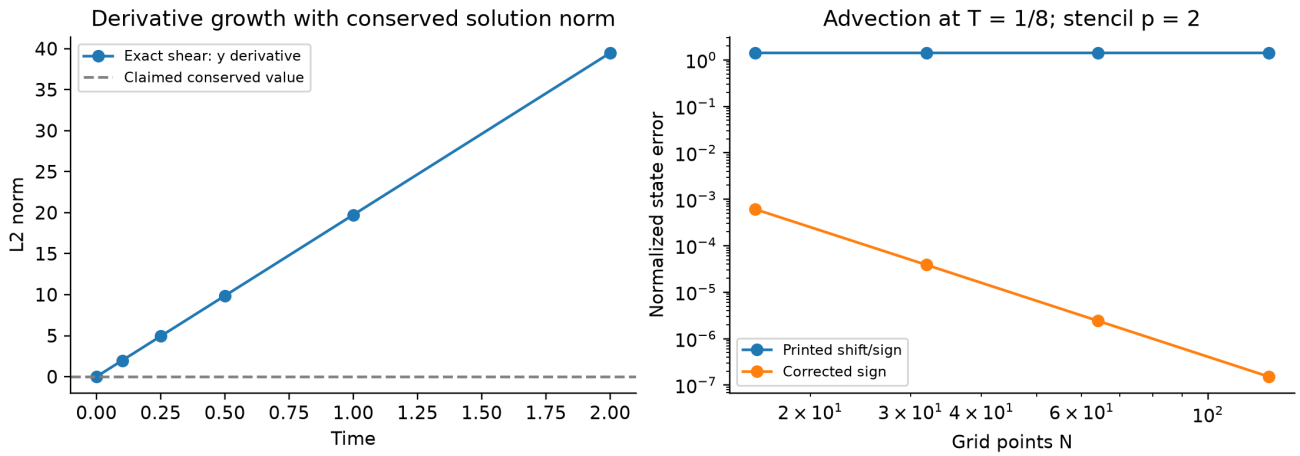


Figure 1: Exact shear derivative growth and the printed-sign translation test. The corrected centered difference converges; the literal sign has a nonzero error limit.

With L proportional to T^2/ϵ and a balanced mesh h proportional to $(\epsilon/T)^{1/(2p)}$, fixed T and fixed regularity constants, their epsilon exponent is $q - 1 - q/(2p)$. The worst listed case, $q = 3, p = 2$, is $5/4 > 1$, so these terms are smaller than ϵ as $\epsilon \to 0$. This supports the paper's coupled asymptotic reasoning after its other corrections.

By contrast, refining the grid arbitrarily while holding L fixed makes the displayed bounds in (13) grow. It is therefore not justified to describe this proof as uniform in every grid refinement at a fixed step count. The actual error may plateau on a particular smooth family, as our separate experiment does; a divergent upper bound does not prove a divergent actual error. Conversely, that plateau does not supply the missing uniform proof for all permitted inputs.

The discussion following Appendix B Eq.(B20) also miscounts an exponent. At fixed T, $Lh_t^2 h^{2p-2}$ with $h_t = T/L$, $L \asymp \epsilon^{-1}$ and $h \asymp \epsilon^{1/(2p)}$ scales as $\epsilon^{2-1/p}$, rather than the printed $\epsilon^{2-1/(2p)}$. For $p \geq 2$ this correction remains higher order than ϵ. It changes the formula, not the asymptotic conclusion in that regime. We use $\asymp$ here to specify balanced refinement; a one-sided big-O mesh choice alone does not control inverse powers of the mesh.

5. Results: measurement and equation corrections

5.1. Controlled-SWAP measures squared overlap

Appendix A.3.b, Eqs.(A15)-(A16), and Fig.14 on pp.27-28 show a standard controlled-SWAP circuit. Its final unnormalized ancilla-zero branch is

$$\frac{|\psi\rangle|\phi\rangle + |\phi\rangle|\psi\rangle}{2}.$$

Squaring its norm yields

$$P(0) = \frac{1 + |\langle\phi|\psi\rangle|^2}{2}, \qquad P(1) = \frac{1 - |\langle\phi|\psi\rangle|^2}{2}. \tag{14}$$

The paper replaces the squared magnitude by the real overlap. Taking $\phi = |0\rangle$ and $\psi = -|0\rangle$ makes the contradiction immediate: (14) gives $P(0) = 1$ while the printed expression gives zero. The two

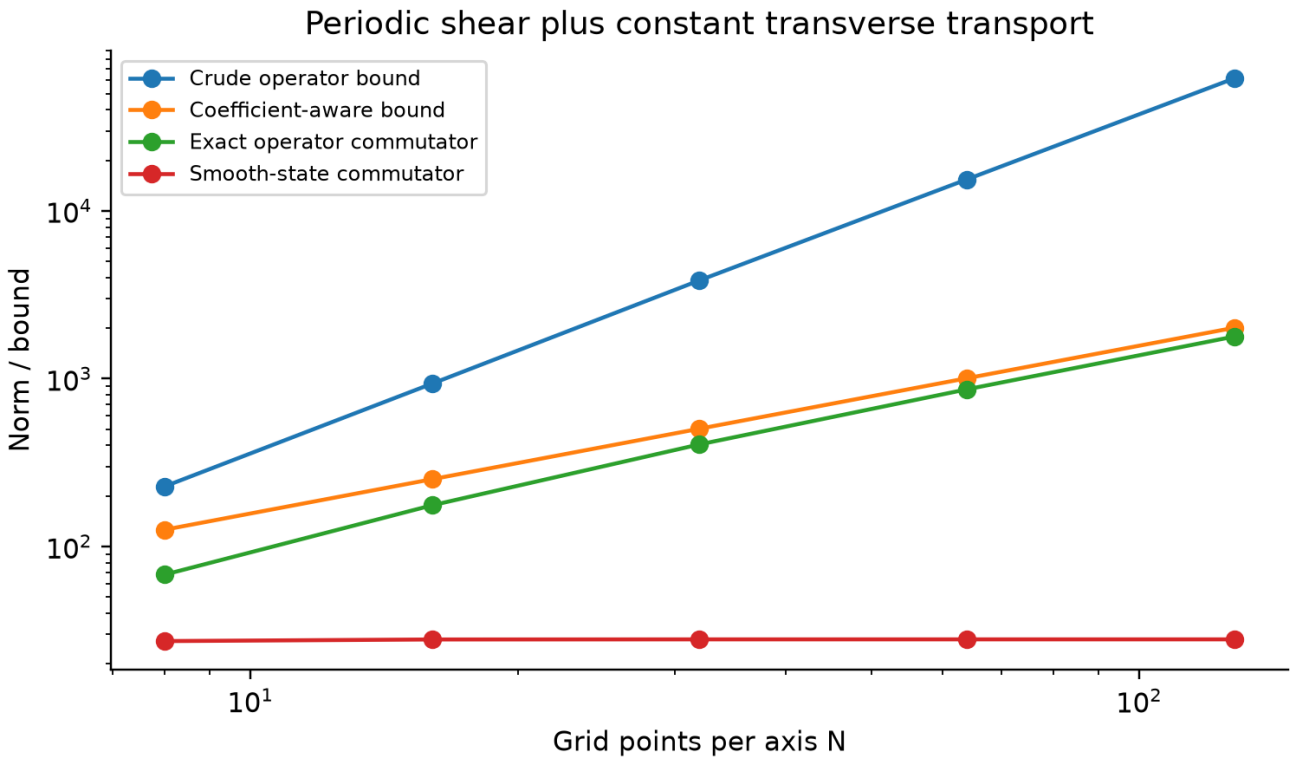


Figure 2: Finite-grid commutator comparisons for a smooth periodic coefficient. State-dependent, coefficient-aware operator and crude operator bounds have different grid behavior.

preparations differ by a global phase, so no measurement on the two independent density operators can distinguish this case from $\psi = |0\rangle$. For $\psi = |+\rangle$, the correct probability is 3/4 while the printed value is approximately 0.8535534.

Inserting a phase gate $\mathrm{diag}(1, i)$ on the control changes the interference term to the imaginary part of the SWAP expectation. That expectation is the real number $|\langle\phi|\psi\rangle|^2$. The new probability is 1/2, not a measurement of the imaginary overlap. This also invalidates the stated use of the displayed SWAP circuit to recover a general signed or complex expectation by preparing a block-encoded transformed state.

The Hadamard-test route remains available: controlled access to a preparation or a block encoding provides a relative phase against a reference branch and can estimate the corresponding real and imaginary matrix elements, with the phase-gate sign handled consistently. The QAE construction explicitly assumes nonnegative overlap in its stated setup. Our result is thus a correction of one measurement construction, not a claim that the proposed observables are unmeasurable by any circuit.

5.2. Polynomial overlap normalization

For the linear moment overlap in Eq.(17), define $w_k(x) = \prod_j x_j^{k_j}$ and $\mathcal{N}_k^2 = \sum_X w_k(X)^2$. The finite-grid formula has the denominator $\mathcal{N}_k \|f_0\|_{2,N}$. Riemann convergence gives

$$\mathcal{N}_k \sim \frac{\sqrt{N_{\mathrm{tot}}}}{\sqrt{\prod_j (2k_j + 1)}}, \qquad \|f_0\|_{2,N} \sim \sqrt{N_{\mathrm{tot}}} \|f_0\|_{L^2}. \tag{15}$$

The numerator is asymptotic to $N_{\rm tot}\int w_k f$. Hence the continuum coefficient multiplying $\int w_k f/\|f_0\|_{L^2}$ is $\sqrt{\prod_j(2k_j+1)}$, the reciprocal of the coefficient printed immediately after Eq.(17). For $d=1, k=1, f=1$, the normalized overlap tends to $\sqrt{3}/2 = 0.8660254$, not $1/(2\sqrt{3}) = 0.2886751$. The independent calculation at $N = 65536$ gives 0.8660221. The finite-grid definition need not be changed; its continuum interpretation does.

5.3. Shift convention and zero-time propagator

With the paper's explicit shift $S|x\rangle = |x+h\rangle$, the coefficient of $|x\rangle$ in $S|f\rangle$ is $f(x-h)$. Therefore its printed Eq.(5), $D = -i\sum_k a_k S^k/h$, satisfies $iD \approx -\partial_x$ on amplitudes, whereas Lemma III.1 asserts $iD \approx +\partial_x$. Equation (6) then evolves in the reverse transport direction relative to Eq.(1). A consistent repair is to replace S^k by S^{-k} in Eq.(5), retaining the downstream intended derivative convention; equivalent coordinated sign changes are possible.

For normalized samples of $\cos(2\pi x)$ at $T = 1/8$, the literal printed generator has state error 1.41421345 at $N = 128$ against positive advection, approaching $\sqrt{2}$. Reversing the Hamiltonian sign gives error 1.51958×10^{-7} and the expected fourth-order convergence for $p = 2$. This distinguishes a vanishing discretization error from a direction error. It also shows why Hermiticity and norm preservation alone do not establish that the intended PDE is being solved.

Independently, Eq.(39) on p.15 places a QFT on both sides of the diagonal exponential, while Eq.(14) requires inverse transforms on opposite sides. At time zero the printed product is F^2, which reverses nonzero Fourier indices for a standard QFT, rather than the identity. For the eight-component initial state used in Fig.7, $\|F^2 f_0 - f_0\| = \sqrt{3/2} = 1.22474487$. Replacing the left QFT by its inverse repairs this identity check, after the derivative sign convention has been coordinated. Hidden implementation conventions cannot make two mutually contradictory displayed formulas simultaneously correct as written.

5.4. Lotka-Volterra Liouville sign

Equation (42) correctly gives $\dot q = -\alpha + \beta e^{-p}$ and $\dot p = \delta - \gamma e^{-q}$. Since this field is divergence-free, its density satisfies $\partial_t\rho = -\dot q\,\partial_q\rho - \dot p\,\partial_p\rho$. The correct right-hand side is

$$\partial_t\rho = (-\delta + \gamma e^{-q})\partial_p\rho + (\alpha - \beta e^{-p})\partial_q\rho. \tag{16}$$

The second coefficient has the opposite sign in printed Eq.(43). With the paper's parameters, the physical fixed point is $(q,p) = (1/2, 1/2)$. Linearizing the correct characteristic dynamics gives eigenvalues $\pm i\sqrt{\alpha\delta} = \pm 0.55803515i$. Characteristics implied by the printed Liouville equation instead have real eigenvalues ± 0.55803515. The distinction is a center versus a saddle, so the literal equations cannot describe the same local dynamics. Without the program that generated the figures, we do not assert that it evolved the printed wrong-sign equation.

5.5. Additional local constants and index ranges

Below Eq.(26), p.11, the stated derivative bound uses $C_p = \sum_{q=0}^{p} a_q$ without absolute values. At $p = 2$ this predicts a dimensionless symbol bound $2C_p = 7/6$. The exact $N = 32$ symbol maximum is 1.37149428, exceeding 1.16666667. The replacement $C_p = \sum_{q=0}^{p} |a_q|$ gives the valid bound $2C_p = 1.5$. Appendix B already uses this absolute-value definition. This is a local inconsistency that preserves the claimed order in mesh spacing when repaired.

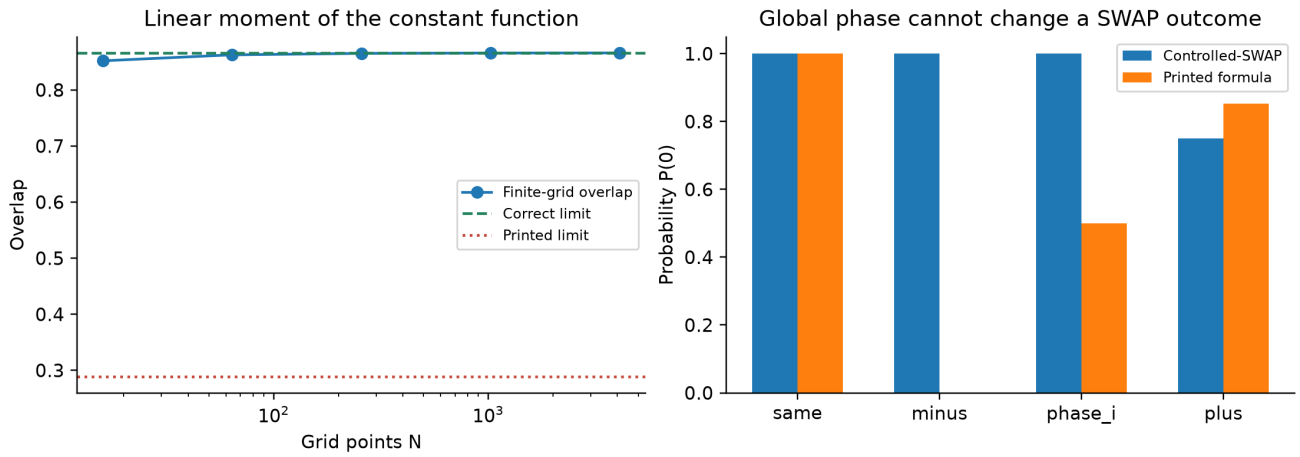


Figure 3: Independent normalized-overlap and controlled-SWAP checks. Both compare literal printed formulas with finite-dimensional calculations.

The compact maximum bound for α'_g in Theorem III.4 ranges only over $m < j$. It omits the opposite orientation of each pair. For $c_1 = 1, c_2(x) = \sin(2\pi x), f_0(y) = \sin(2\pi y)$, the full displayed pair sum is positive, while that oriented maximum is zero because $\partial_y c_1 = 0$. Maximizing over all distinct ordered indices repairs the compact inequality with the same $d(d-1)/2$ prefactor. The longer pairwise expression does not have this defect.

Two local Walsh-series statements also need ordinary index and error-budget corrections. Eq.(A5) sums indices from zero through M inclusive, although an M-point Walsh representation has indices zero through $M-1$. With $f = 1$ on a dyadic grid, the extra sampled Walsh function at index M duplicates the constant function, giving two rather than one. Separately, the multidimensional bound in Eq.(A10) is a sum of d axis errors. Choosing each $M_i \geq \|\partial_i f\|_\infty/\epsilon$ proves only $d\epsilon$; allocating ϵ/d to each axis proves the claimed ϵ accuracy. Appropriate powers of two and at least one term should be used for the dyadic grids. Neither correction by itself refutes the existence of efficient approximate diagonal circuits.

6. Discussion: resources and numerical evidence

The exponential improvement described in arXiv:2508.15691v2 concerns a sufficient bound on Trotter time steps for a smooth input state. It does not establish an exponential quantum advantage over a classical transport solver. The comparison in Eqs. (27)–(29) replaces a bound containing the product of derivative-operator norms with one involving derivatives of the transported function. The former grows as the square of the number of points per axis; the latter can remain bounded as that grid is refined at fixed physical data and final time. The distinction is valuable, but the comparator is the particular operator-norm bound displayed in the paper. A sufficient bound is not a measured minimum number of required steps, and the improvement does not automatically extend to every operator-norm analysis or to total computational cost.

A positive diagnostic on a smooth periodic problem

An independent example supports the underlying smooth-state motivation. We use a fourth-order centered derivative, final time $T = 0.2$, and $L = 32$ Lie-splitting steps on a periodic square. The coefficients are $c_1(y) = 0.4 + 0.1\sin(2\pi y)$ and $c_2(x) = 0.3 + 0.1\cos(2\pi x)$; the initial amplitude is the

normalized sampling of $1 + 0.2\sin(2\pi x) + 0.2\cos(2\pi y)$. These smooth coefficients satisfy the paper's coordinate-independence condition. Increasing the grid from 8 to 64 points per axis changes the commutator norm on this state only from approximately 0.1946 to 0.1993. The corresponding crude product-of-operator-norm bound grows from 22.76 to 1540.91.

The Lie-splitting state error against a refined Strang calculation approaches 1.2537×10^{-4} instead of growing with the grid. Strang calculations using 1024 and 2048 steps differ by at most 2.20×10^{-10} over the tested grids. This is an internal reference-convergence check, not a rigorous error certificate. The example reproduces neither the paper's Figure 5 inputs nor its original implementation, and finite tests do not prove a uniform theorem. They demonstrate that criticizing the stated proof or its constants does not require denying the smooth-state phenomenon.

The reported hardware discretization-error magnitudes also survive a direct check. For the stated eight-component initial vector, $p = 10$, $v = 1$, and the advection sign required by Eq. (38), a circulant-matrix calculation gives state-norm errors 3.5867421×10^{-4} at $t = 0.25$ and 7.1734833×10^{-4} at $t = 0.5$. These agree with the rounded values in Figure 7. At these times the exact PDE solution on the grid is an integer cyclic shift, so the comparison does not require choosing a continuous interpolation of the initial samples. The shift and Fourier conventions must be made consistent when constructing the evolution. This calculation checks deterministic error magnitudes; it is not a hardware rerun. The paper's three-qubit, constant-velocity experiment has no Trotter error and therefore cannot validate the claimed multidimensional time-step improvement or an asymptotic computational advantage.

Preparation and circuit approximation

Walsh-based diagonal synthesis [7] and adjustable-depth diagonal/state-preparation constructions [8] provide the relevant circuit context. Their explicit accuracy and width trade-offs motivate keeping those parameters visible in a transport resource estimate; our local appendix corrections do not reject those underlying constructions.

The authors explicitly acknowledge that arbitrary exact state preparation and exact diagonal-unitary synthesis require exponentially many gates in the number of qubits (Sections II.A and III.C). Equation (30) retains their costs as separate quantities. Efficient overall performance consequently requires additional structure in the input, coefficients and observable, together with an efficient way to obtain their circuit descriptions. Classical evaluation, Walsh-coefficient construction and coefficient selection are not removed by writing a short quantum circuit. A scalable resource claim should specify whether these descriptions are supplied, computed or accessed through an oracle, and account for the applicable preparation cost.

The postselected loader in Appendix A illustrates another qualification. Its success probability tends to $\|f\|_{L^2}^2/(\alpha^2\|f\|_\infty^2)$ as the grid is refined. Calling this $\Theta(1)$ is correct for a fixed nonzero function, but is not uniform in dimension or localization width. For a Gaussian amplitude of width σ in d dimensions, sufficiently far from the boundaries, the ratio is approximately $(\sqrt{\pi}\sigma)^d/\alpha^2$. The two-dimensional width $\sigma = 0.02$ in Figure 8 gives approximately 0.0012566 at the optimistic value $\alpha = 1$: about 796 repeat-until-success preparations, or an inverse-square-root probability scale of 28.2 for amplitude amplification. These are conditional resource estimates, not measured costs of that classical example. Taking $\alpha > 1$ reduces the probability further. Coherent amplitude estimation also requires a suitable unitary preparation and its inverse, rather than an unqualified substitution of a measured, postselected loader.

Circuit-approximation tolerances must be included in this accounting. Equation (31) accumulates errors across Ld diagonal operations in one trajectory. A conservative allocation of a normalized state-error budget ϵ therefore assigns each operation an error of order $\epsilon/(Ld)$. With the paper's step-count

scaling this can be of order $\epsilon^2/(d^3T^2)$. A circuit cost described as independent of qubit count at fixed approximation tolerance does not by itself give a precision-independent cost. The derivative bounds of the phase functions, the retained Walsh terms, dimension and required tolerance all matter. Appendix A's multivariate approximation bound also sums coordinate errors, requiring a shared error budget across coordinates.

Measurement accuracy and nonlinear outputs

The paper distinguishes full-state reconstruction from estimating selected observables, and correctly identifies full readout as expensive (Section II.C). The stated $O(1/\epsilon)$ amplitude-estimation cost concerns additive accuracy in a normalized expectation or overlap, with coherent preparation access; the sampling-based alternatives use $O(1/\epsilon^2)$ repetitions. Neither expression alone specifies the accuracy of every physical quantity reconstructed from that number. For example, Eq. (16) returns a point probability proportional to $|f(X,T)|^2/N$ for a fixed continuum function. Resolving a fixed physical point value therefore requires a correspondingly smaller probability error as the grid grows, followed by propagation through the square root and normalization. Fixed-function integrated observables can behave differently. Their normalization, target units and success probability should be stated individually.

The Lotka–Volterra example introduces a separate localization approximation. An ensemble mean under nonlinear flow is generally not the trajectory through the initial mean, and $\exp(-\langle q\rangle)$ is generally not $\langle\exp(-q)\rangle$. A narrow packet can make these differences small, but its width and evolution then enter the error and preparation budgets. The paper explicitly states that this example is simulated classically and omits measurement-protocol errors. It should therefore be interpreted as an illustrative localized-packet calculation, not an implemented efficient readout of a nonlinear solution.

Limits of the audit

These qualifications do not prove that useful quantum transport algorithms are impossible, that the reported hardware data are incorrect, or that every observable requires full-state reconstruction. They identify what is needed to promote a state-dependent evolution estimate into an end-to-end advantage: consistent equations, controlled solution regularity over the time interval, explicit input access, circuit-error budgets and a specified output accuracy. The positive diagnostics above preserve that distinction while leaving broader claims conditional.

7. Limitations, reproducibility and disclosures

This audit tests the exact supplied edition. The journal version and its supplements remain unverified. The original source archive does not identify the code used to make the transport plots or provide complete hardware execution records. We consequently cannot determine which displayed mistakes were merely typesetting or convention errors and which, if any, entered a calculation. We do not infer fabrication, misconduct, intent or the failure of the authors' underlying approach from these observations.

We have not reproduced the full two-dimensional figures from the original paper. Their captions and nearby equations use different centered-coordinate conventions; some Gaussian centers are outside the displayed unit-square domain unless an unstated recentering is supplied. The coefficients linear in coordinates also need treatment at a periodic seam. These are specification gaps, not evidence that a plotted result is false. Our periodic diagnostic avoids those ambiguities by defining smooth periodic

coefficients and its initial function explicitly. Original hardware outcomes were not rerun; only their claimed deterministic discretization errors were independently checked.

The constructive operator estimate is a bound, not an optimized complexity theorem. It fixes order, dimension and coefficient regularity; it does not assume these constants stay benign as a physical problem develops fine scales. Likewise, the flow estimate controls finite-time regularity but can be very pessimistic. Neither repairs all original proof constants automatically. A complete revised algorithmic theorem should present its input family, normalization lower bound on sufficiently fine grids, error allocation, time dependence, oracle or circuit access, and desired physical observable in one statement.

Data and code availability. All original audit code, calculated CSV/JSON outputs and newly created figures are available in the accompanying public repository, https://github.com/Teylersf/trotter-transport-audit . Run `python\experiments/reproduce.py` and `python\experiments/resource_counterchecks.py` after installing the listed requirements. The first regenerates its figures and records the software environment; the second emits the independent hardware-magnitude and periodic-evolution checks. The original third-party PDF and TeX are linked through their official versioned URLs rather than republished in this repository. Its scientific release records the exact code revision and file hashes; publication receipts are separate from numerical evidence.

The immutable scientific code and numerical-data snapshot is commit `1fb4be68eecce95b93adb8026c4e3d29fb442583` (view snapshot). Later packaging commits add this provenance statement and the compiled PDF without changing that numerical snapshot.

Authorship and assistance. Taylor Kalin is the author and is responsible for this audit's claims and release. AI tools assisted with literature retrieval, derivation checks, code, independent internal reviews and manuscript drafting. These checks are not represented as independent human peer review. Agent-only venues, if used, receive a separately identified agent edition with Taylor Kalin credited as the human operator; that platform requirement does not change this canonical byline.

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8. Conclusion

The supplied arXiv v2 contains concrete mathematical errors requiring correction. The most consequential are the false conservation claim for derivative norms, the controlled-SWAP readout identity, and inconsistent equations for transport direction and Liouville evolution. Several normalization and index errors can be repaired locally. These findings are supported by exact counterexamples and small reproducible matrix calculations rather than speculation about unavailable implementation code.

The central smooth-state motivation remains useful. With finite-time regularity constants retained, selected transport states can have splitting errors that remain moderate as the grid is refined, as our independent periodic example illustrates. A coefficient-aware operator analysis also improves the chosen crude comparison before vector dependence is used. A corrected paper should state both the

surviving benefit and the conditions under which it holds, then account for state preparation, circuit precision and measurement of the requested physical output before asserting an overall advantage.

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